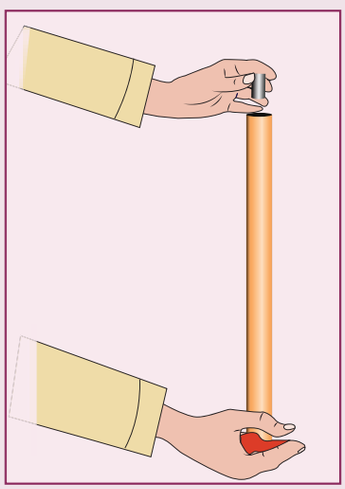


Solution

From right hand rule, the magnetic field by the straight wire is directed into the plane of the square loop perpendicularly and its magnetic flux is decreasing. The decrease in flux is opposed by the current induced in the loop by producing a magnetic field in the same direction as the magnetic field of the wire. Again from right hand rule, for this inward magnetic field, the direction of the induced current in the loop is clockwise.

Activity

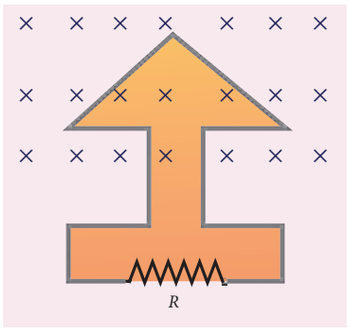


Demonstration of Lenz's law

Take a narrow copper pipe and a strongly magnetized button magnet as shown in figure. Keep the copper pipe vertical and drop the magnet into the pipe. Watch the motion of the magnet and note that magnet has become slower than its free fall. The reason is that an electric current generated by a moving magnet will always *oppose* the original motion of the magnet that produced the current.

EXAMPLE 4.7

The magnetic flux passes perpendicular to the plane of the circuit and is directed into the paper. If the magnetic flux varies with respect to time as per the following relation: $\Phi_B = \left(2t^3 + 3t^2 + 8t + 5\right)\text{mWb}$, what is the magnitude of the induced emf in the loop when $t = 3$ s? Find out the direction of current through the circuit.



Solution

$\Phi_B = \left(2t^3 + 3t^2 + 8t + 5\right)\text{mWb}$; $N = 1$; $t = 3$ s

(i) $$\varepsilon = \frac{d\left(N\Phi_B\right)}{dt}$$

$$= \frac{d}{dt}\left(2t^3 + 3t^2 + 8t + 5\right)\times 10^{-3}$$

$$= \left(6t^2 + 6t + 8\right)\times 10^{-3}$$

At t = 3 s,

$$\varepsilon = \left[\left(6\times 9\right) + \left(6\times 3\right) + 8\right]\times 10^{-3}$$

$$= 80\times 10^{-3}\,\text{V} \quad = 80\,\text{mV}$$

(ii) As time passes, the magnetic flux linked with the loop increases. According to Lenz's law, the direction of the induced current should be in a way so as to oppose the flux increase. So, the induced current flows in such a way to produce a magnetic field opposite to the given field. This magnetic field is perpendicularly outwards. Therefore, the induced current flows in anti-clockwise direction.

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